Lesson 5.2 **Duration: 40 minutes**

Flowcharts

Learning Objectives

- Interpret and construct flowcharts using standard symbols
- Map algorithmic processes to flowchart representations
- Evaluate when flowcharts are more effective than pseudocode for communication

Engage (5-7 minutes)

Show a flowchart of a common decision process (e.g., “Should I bring an umbrella?”) without explaining the symbols. Ask students to trace the logic. Then reveal the standard symbols and discuss how visual representation clarifies complex decision-making compared to plain text.

Explore (10-12 minutes)

Using paper and stencils or a digital tool, pairs create flowcharts for a login authentication process: entering a username/password, checking against a database, granting or denying access, and handling incorrect attempts. Students must use correct symbols: ovals for start/end, rectangles for processes, diamonds for decisions, and parallelograms for input/output.

Explain (10-12 minutes)

Flowcharts are graphical representations of algorithms using standardized symbols connected by arrows showing control flow. The main symbols are: terminal (oval) for start/stop, process (rectangle) for computations, decision (diamond) for conditional branches, input/output (parallelogram) for data operations, and flowlines (arrows) for sequence direction. Flowcharts make complex logic visible and are excellent for communicating with non-technical stakeholders.

Elaborate (8-10 minutes)

Analyze flowcharts for real systems: ATM withdrawal process, online order checkout, or a traffic light controller. Discuss advantages (visual clarity, universal understanding) and limitations (difficult to maintain for large systems, imprecise for implementation). Compare a flowchart side-by-side with pseudocode for the same algorithm and discuss which is better for different audiences.

Evaluate (5-8 minutes)

Students create a flowchart for a given algorithm (e.g., finding the largest of three numbers) and peer-review for correct symbol usage, logical flow, and completeness. Quiz: identify errors in a deliberately flawed flowchart (missing branches, wrong symbols, infinite loops).